

Summary

AI/ML Engineer with experience building agentic AI systems, multilingual conversational AI platforms, and ML training pipelines. Skilled in Python, Java, PyTorch, Hugging Face, LangChain, LangGraph, and backend systems including Snowflake, Airflow, PostgreSQL, FastAPI, and REST APIs. Experienced in LLM fine-tuning, RAG pipelines, speech AI, multimodal AI, and end-to-end deployment of production-grade AI workflows. Focused on building scalable, reliable, and real-time AI-powered systems across GenAI and conversational AI applications.

Experience

Worldish

Software Engineer

Dec 2025 - Present

- Took technical leadership in AI/ML-focused projects and feature development, driving experimentation, architecture decisions, and production deployment of multilingual AI systems.
- Created MLOps pipelines for rare and low-resource language fine-tuning workflows, enabling scalable dataset preparation, model training, evaluation, and deployment for multilingual AI systems.
- Redesigned core AI product pipelines to improve multilingual orchestration, inference scalability, and cross-service reliability for production healthcare AI workloads.
- Engineered multilingual conversational AI systems for healthcare applications supporting low-resource languages including Tigrinya, Wolof, Bambara, and Arabic dialects.
- Built an Arabic dialect AI evaluation platform integrating LLM prompting, context engineering, Speech-to-Text (STT), and Text-to-Speech (TTS) workflows across multiple AI providers.
- Integrated and benchmarked multiple LLM, STT, and TTS providers based on multilingual accuracy, latency, and conversational quality for real-time AI interactions.
- Developed real-time multilingual voice AI pipelines combining translation, speech recognition, and neural speech synthesis for healthcare conversational systems.

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Member of Technical Staff 2

Jan 2025 - Dec 2025

- Engineered an agentic AI framework (Langchain & LangGraph) to automatically triage Apache Airflow pipeline failures combining rule-based logic with LLM-driven classification, Slack alerting, and safe auto-remediation.
- Designed and deployed LLM-based log summarization (LangChain + Ollama) that converts raw Airflow logs into concise developer-facing insights, reducing debugging time and improving alert clarity.
- Built an AI-driven anomaly detection service for ETL execution metadata (DAG/task states, XComs, SIGTERM patterns) and surfaced pipeline bottlenecks via PostgreSQL-backed dashboards used by SREs and platform teams.
- Developed a real-time incident response pipeline where ML classifiers auto-route alerts to Slack and auto-create Jira tickets cutting MTTR by ~60% and standardizing on-call workflows.
- Extended orchestration for petabyte-scale ETL across AWS & GCP; modularized data pipelines to reliably feed ML training datasets.
- Participated in supervised fine-tuning of Azure OpenAI models to adapt it for ThoughtSpot internal workflows. Wrote training loops, dataset preprocessing, and evaluation scripts to iterate on models (PyTorch).

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AI/ML Intern

Jun 2024 - Dec 2024

- Built backend services & robust data pipelines for pricing and metrics datasets, enabling downstream ML workflows.
- Developed an initial RAG prototype (LangChain + Haystack) that indexed pricing/metrics into vector stores and supported contextual Q&A, experimented with embedding models to improve retrieval relevance.
- Prototyped agentic workflows where lightweight AI components (retrieval, summarization, alerting) cooperatively automated repetitive analysis tasks.

Key Projects

Agentic AI System for Data Pipeline Reliability

- Built production-scale application of agentic AI by deploying agentic workflows in LangGraph that autonomously handled triage, log summarization, root cause analysis, and remediation—cutting MTTR by ~60%.
- Enhanced reliability and accuracy through LLM reasoning with retrieval-augmented context (vector search over historical incidents) and guardrails for safe action execution in production.

MediScan: Precision Medical Record Digitization with OCR and LSTM

- Integrated Google OCR with advanced preprocessing to enhance accuracy in digitizing medical records, leveraging LSTM models and terminology correction.
- Developed a Flask-based interface for seamless image-to-text conversion.

Sentiment Analysis Application Using BERT

- Developed a BERT-based sentiment analysis model, deployed a REST API for real-time classification.
- Enhanced model accuracy for minority classes through oversampling, enabling more balanced and data-driven insights.

VisioVoice: Multimodal Conversational AI for Accessibility

- Built a multimodal conversational AI system enabling blind-first voice interaction using one-shot vision analysis, conversational memory, and real-time Speech-to-Text (STT) / Text-to-Speech (TTS) workflows.
- Designed a grounded AI architecture where images were analyzed once into structured JSON representations, enabling deterministic reasoning and conversational follow-up without re-sending images to the model.
- Developed real-time voice interaction pipelines using FastAPI, React, Azure OpenAI Vision, and Azure Speech Services supporting continuous conversational loops, hazard-aware narration, and spatial scene understanding for accessibility-focused AI interactions.

Skills Summary

• Programming Languages	Python, Java, SQL, JavaScript, C.
• Generative AI & Machine Learning	Agentic Workflows, LangChain, LangGraph, Prompt Engineering, Context Engineering, RAG, Vector embeddings, LLM Fine-tuning, AI Evaluation & Benchmarking, Multilingual NLP, Speech AI (STT/TTS), Voice AI Systems, PyTorch, Hugging Face(Transformers), Google OCR, Conversational AI, TensorFlow.
• Data & Analytics	Airflow, Data Pipelines, Snowflake, Feature Engineering, Structured/unstructured data ingestion, preprocessing.
• MLOps	Grafana, Prometheus, MLflow, Data Preparation, Ollama.
• Tools and Frameworks	Django, Flask, FastAPI, Pydantic, MongoDB, React, NodeJS, Git, GitHub, VS Code,Cursor, Claude Code, OpenAI Codex, Gemini Code Assist, API Testing & Development (Postman), AWS, Azure, GCP.

Education and Scholastic Achievements

Jawaharlal Nehru Technological University, Anantapur	India
Bachelor of Technology - Computer Science and Engineering & Business Systems; GPA 7.6/10	Feb 2020 - May 2024
• Secured Rank 1973 in CodeKaze Coding Competition among 200k+ participants	
• Elite Certificate in Internet of Things (IoT) from NPTEL	
• Participated in few impactful hackathons and prototyped innovative ideas including in field of NLP and AI	

Publications

- **Mr.B. Varun Kumar Reddy**, Dr.G. Kishor Kumar, Mr.S. Sohel, Mr.G. Sai Kumar, Mr.N. Vivek Naidu, Mr.K. Sudheer Reddy. **Digitization of Medical Records using OCR**. International Journal of Advanced Research in Computer and Communication Engineering. DOI: 10.17148/IJARCCE.2024.134141.

Certifications

Java Programming and Software Engineering Fundamentals Specialization by Duke University

- Comprehensive implementation of an information retrieval and recommendation system.
- Proficient use of data structures, design patterns, database management, and Java programming expertise.

Machine Learning Specialization by DeepLearning.AI and Stanford University

- Gained expertise in applying both supervised and unsupervised machine learning models utilizing tools such as NumPy, scikit-learn, and TensorFlow.
- Cultivated abilities in constructing regression, classification, clustering, recommender systems, and deep reinforcement learning models.

Generative AI with Large Language Models

- Mastered LLM-based GENAI lifecycle: model selection, fine-tuning, deployment.
- Proficient in transformer architecture and optimizing LLM performance for applications.

AWS Certified Data Engineer – Associate (DEA-C01)

- Validated expertise in designing, building, and optimizing scalable ETL/ELT data pipelines on AWS.
- Hands-on expertise in data lake/warehouse architectures (S3, Glue, Redshift, EMR, Athena) for analytics and ML workflows.
- Applied AWS best practices for pipeline reliability, monitoring, and security in production environments.

Positions of Responsibility

Class Representative , CSE & BS Department	2020
Core Member and Contributor , Tech Community	2022
Team Member and Student Coordinator , ISTE Chapter	2022

Achievements/ Extracurricular Activities

- Led the basketball team to the South Zone league finals at university level.
- Organized and took part in various intercollegiate Hackathons and Designathons.